

FACULTY TRAINING

AI for Research

Prompt-based data analysis with Claude and ChatGPT, done ethically

<https://rifat963.github.io/ewu-ai-training/>

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Materials: github.com/rifat963/ewu-ai-training

TODAY

Five parts, one real dataset, no code to write

1

Ethics first

Concerns, EWU rules, an ethical routine

2

Workflow and prompts

Eleven standard steps; how a prompt is built

3

Live demo and hands-on

3,156 students on AI and career anxiety

4

Prompt to agent

Projects, context files, skills, tools, agents

5

Your ecosystem

How to set up AI for your own research



Follow along: the Prompt Lab website has every prompt with a copy button and the result to expect. Link in the GitHub repository.

WHAT HAS CHANGED

Claude and ChatGPT now run code on your data



You prompt

Plain English, plus an
uploaded Excel or CSV file



AI writes Python

pandas, SciPy,
statsmodels, matplotlib



It runs the code

In a sandbox, on your
actual file



You get results

Tables, charts, clean files
to download

The catch: the code can still be wrong, the test can be the wrong test, and the AI may describe a result it never computed. Always ask to **see** the code, and re-run what matters.

ETHICAL CONCERNS

Six risks every AI-assisted study must manage



Privacy and consent

Participant data sent to an external service; consent may not cover it.



Fabrication

Invented numbers, citations or quotes that look right.



Bias

Skewed samples and AI defaults that favour the majority view.



Reproducibility

Same prompt, different answer; no record of what was done.



Authorship

Undisclosed AI text, plagiarism, AI listed as an author.



Over-reliance

Accepting results you cannot explain or reproduce.

What EWU's ethics framework asks of us

EWU RESEARCH ETHICAL COMMITTEE (EWUREC)

"All research work at EWU should adhere to the highest ethical standards; any research work carried out at EWU which has significant ethical implications will have to be submitted for independent ethical review."

EWU has a Policy and Procedure for Research Ethics Approval and a Plagiarism Policy.

- ✓ **New human-participant data:** submit to EWUREC before collection
- ✓ **Consent** that covers how data will be processed, including by AI tools
- ✓ **Anonymise** before any upload; keep raw data on approved storage
- ✓ **Plagiarism policy** applies to AI-generated text: disclose and cite
- ✓ **Secondary data:** respect the licence and never re-identify

An ethical routine: before, during and after

Before you upload

- Anonymised or synthetic data only
- Licence or permission to use it
- Consent covers this use
- EWUREC status clear
- AI account data settings checked

While you analyse

- Plan before changing data
- See the code for every number
- No re-identification attempts
- No cells under 10 people
- Keep a prompt log

Before you publish

- Recompute key results
- Associational language
- State the limitations
- Cite the dataset
- Disclose AI use

STANDARD DATA ANALYSIS APPROACH

Eleven steps, one standard prompt each



Run the steps in **one conversation** so the AI keeps the context, and keep the human decision at steps 0, 3 and 8.

ANATOMY OF A RESEARCH PROMPT

Role, context, task, format, check

Role	Act as a statistician.
Context	<code>career_anxiety</code> is ordinal with four levels.
Task	Test: H1 anxiety differs by gender; H2 ... Use chi-square with Cramér's V, Spearman's rho, Kruskal-Wallis. Check expected cell counts.
Format	Report a table: hypothesis, test, statistic, df, p, effect size, verdict.
Check	Run the code and show it. Do not use causal words.

The **check** line is the ethical part: it makes the AI show its work and stay within what the data can support.

TYPES OF PROMPTS

Ten patterns, and when to use each

Pattern	What it does	Best for
Zero-shot	A clear task, no examples	Quick facts about the data
Role	Sets whose standards apply	Ethics checks, statistics advice
Context-rich	Role, context, task, format, check	Every reproducible step
Few-shot	Worked examples, then the rest	Coding free-text answers
Plan-then-execute	Plan, approve, then run	Anything that changes data
Structured output	Fixes the table or JSON shape	Results you will compare
Tool use (code)	AI runs Python, shows code	Statistics, charts, files
Self-verification	Recompute and compare	Before numbers leave the chat
Critique	AI argues against your claim	Interpretation, abstracts
Meta-prompt	AI writes a prompt or file	Context files, skills

THE DEMO DATASET

Career anxiety in the age of AI: 3,156 students

3,156

university students

5

universities in Bangladesh

13

variables, two Excel files

CC BY

free reuse with citation

Variables: age, gender, academic year, self-rated AI knowledge, AI tools used, career path, views on AI replacing jobs and on AI's future, and career anxiety (No, Low, Medium, High).

Islam, S., Piyas, B. R. C., Tisha, F. J., Nayem, H. K., Rahman, S., Preenon, B. R. C., & Hossen, S. (2026). Career anxiety in the age of artificial intelligence: Survey data of university students in Bangladesh. *Data in Brief*, 67, 112924. doi:10.1016/j.dib.2026.112924

Ask an ethics reviewer before the first analysis

Act as a research ethics reviewer at a university in Bangladesh... assess licence, consent, re-identification risk, whether uploading to an AI tool is acceptable, and what I must disclose. Answer as a table. Do not analyse the survey answers yet.

Issue	What the file shows	Action
Licence	CC BY 4.0	Cite the article and DOI
Consent	Yes for all 3,156 rows	Report the procedure
Re-identification	Ages 26 and 27: only 5 students each	Group ages; no cells under 10
Original approval	Reported as verbal	State as a limitation

A published dataset still has problems to find

45 missing AI-knowledge answers (1.4%)

6 spellings of "no tool": none, Nothing, None of them...

160 free-text career answers, with typos like "Reseacrher"

1 multi-select column needing a split: "ChatGPT, Quiltbot, Grammarly"



The encoded file has an invalid code: `ai_takeover_time = "h"` for participant 31YD3J. The raw file says "Never", so it should be 0. Recode from text labels.

STEP 3 · CLEAN AND ENCODE · PROMPTS P3, P3B

Plan first, approve, then let the AI execute

P3 · Plan-then-execute

Propose a cleaning plan and wait for my approval before running it... Never drop rows unless I agree.

Result: 3,156 rows in, 3,156 out; ordinal codes; one 0/1 column per AI tool.

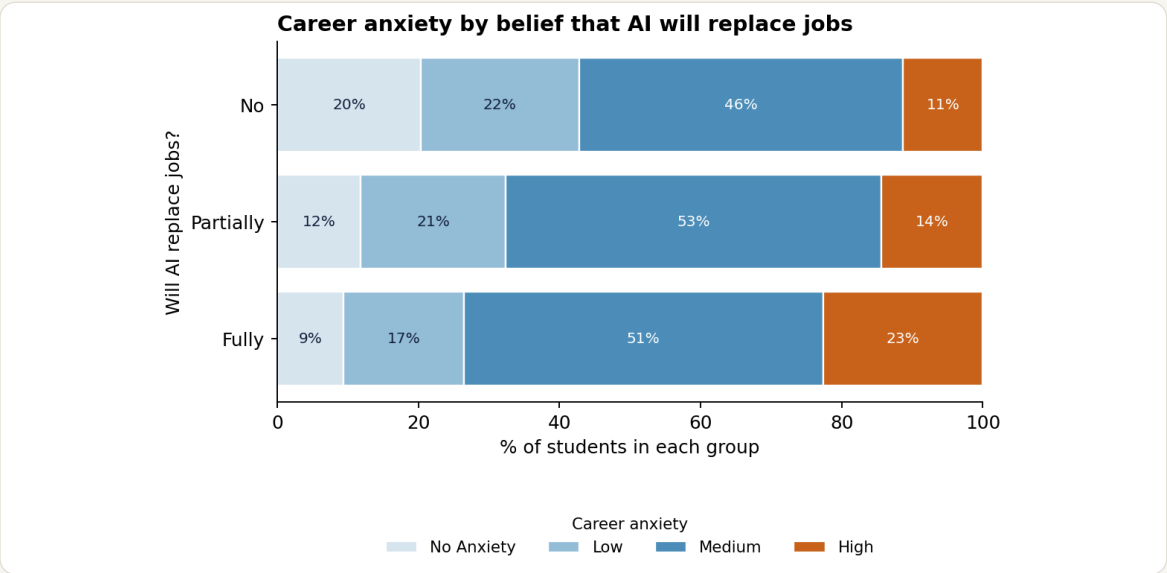
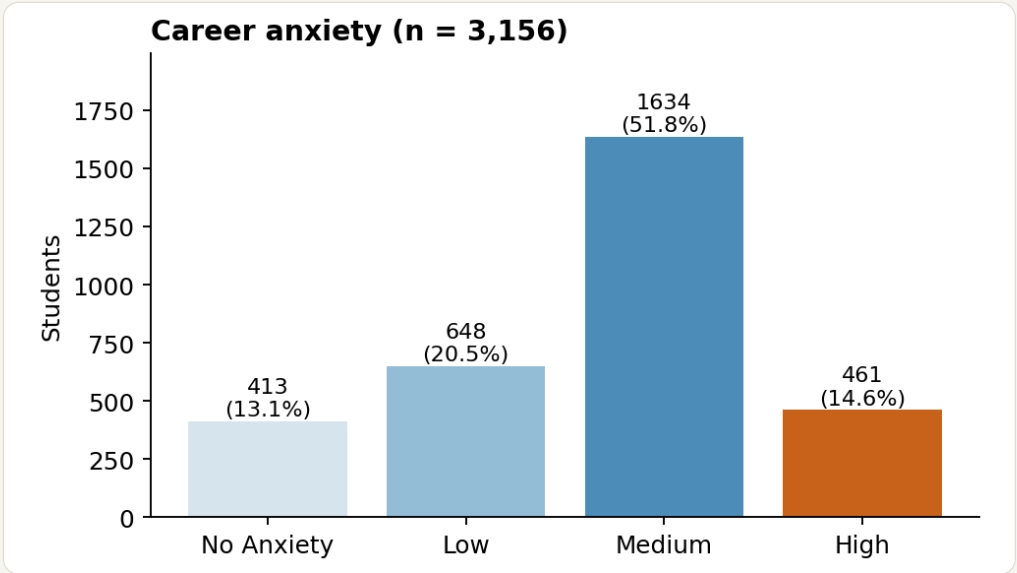
P3b · Few-shot grouping

"Software Engineer" → Software & IT; "Pharmacist" → Health & Food... group all 160 careers.

Result: Software & IT 938, Research & Academia 776, Data & AI 646.

Cleaning decisions are **research decisions**. The AI proposes; you approve; the log goes in your methods.

One prompt, and the AI runs the code for the charts



Two in three students report medium or high anxiety. High anxiety doubles from 11% to 23% between "AI won't replace jobs" and "AI will fully replace jobs".

Significant is not the same as large

Hypothesis	Result	Verdict
H1 Gender	$\chi^2(3) = 6.84, p = .077, V = .05$	No difference
H2 AI knowledge	$\chi^2(6) = 30.79, p < .001, V = .07$	Very small
H2 trend	$\rho = -.02, p = .271$	No trend
H3 Replace jobs	$\chi^2(6) = 51.59, p < .001, V = .09$	Small
H4 Academic year	$H(3) = 5.00, p = .171$	No difference

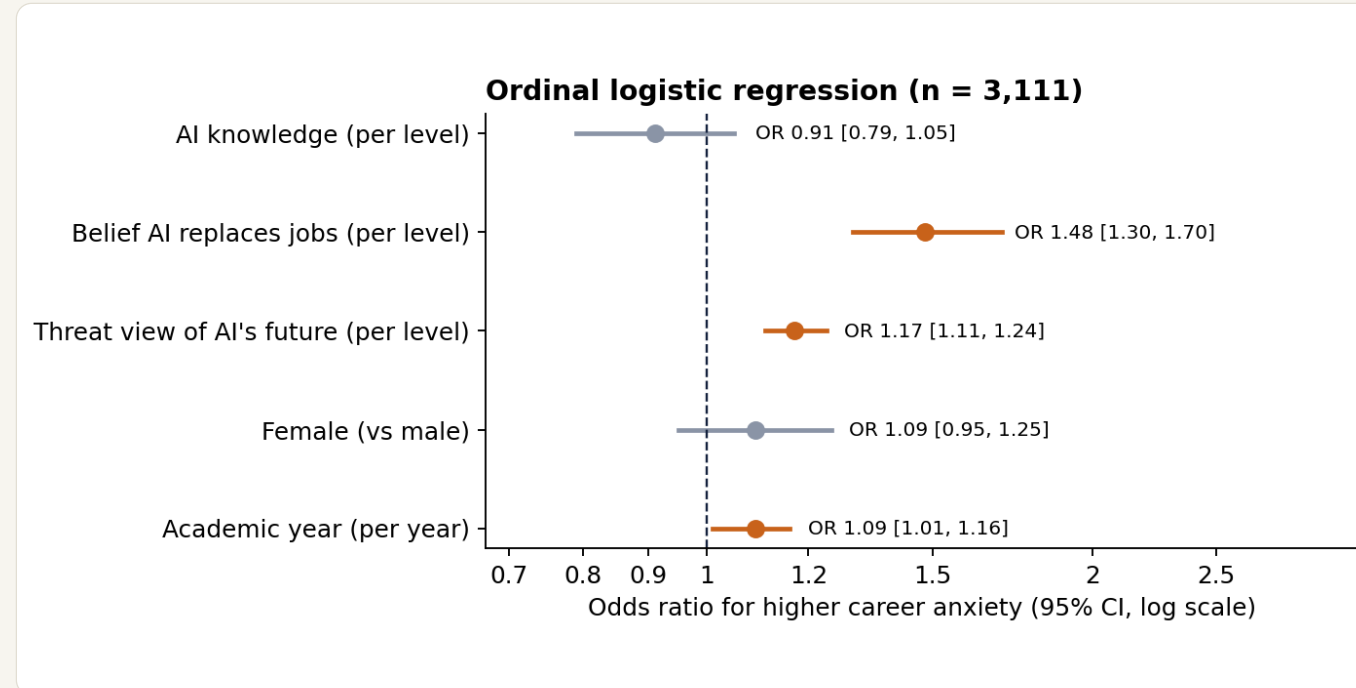
Ask the AI why

H2 is significant but has no trend. Students with high AI knowledge are the most likely to report **no** anxiety (22%) and also **high** anxiety (18%).

Every effect is small (Cramér's $V \leq .09$). With 3,156 students, small differences reach $p < .05$.

STEP 7 • MODEL • PROMPT P7

Belief that AI replaces jobs is the clearest predictor



1.48×

odds of higher anxiety per step in believing AI will replace jobs

$R^2 = .011$

McFadden's pseudo- R^2 : these views explain very little of anxiety

Make the AI check itself, then argue against you

PV • Self-verification

Re-compute every number in your last answer from the raw file in a fresh code cell... show reported vs recomputed.

A mismatch means a different cleaning step or silently dropped rows. Stop and find out.

P8 • Sceptical reviewer

- Single-item, self-reported anxiety
- Cross-sectional: no causal direction
- Five universities, mostly technical students
- AI seminar just before the survey
- Binary gender; verbal ethics approval

Cite the data, say what the AI did, own the result

AI-use disclosure

"[Tool and version] was used to write and run the analysis code and to draft this paragraph from our prompts. All numbers were verified by the authors by re-running the code on the original file, and the authors take full responsibility for the content."

✓ Dataset cited with DOI

✓ No AI as author

✓ Prompts and code shared

YOUR TURN

Download, prompt, compare, verify

1 • Download

dataset.xlsx from the Prompt Lab

2 • Tick the checklist

Ethics before upload

3 • Prompt in order

P0 to P10 in Claude or ChatGPT

4 • Compare

Your numbers vs the possible result

github.com/rifat963/ewu-ai-training

MODERN AGENTIC SYSTEMS

From a prompt to an agent, in seven levels



Each level keeps the one below and adds memory, procedure or action. **More autonomy needs more verification.**

USAGE

Which layer for which job

Layer	Holds	Loaded	In Claude / ChatGPT
Project	Instructions and files for one study	Every chat in the project	Claude Projects · ChatGPT Projects
Context file	Facts and rules in Markdown	Start of every chat or session	In a Project; CLAUDE.md in Claude Code; AGENTS.md for many agents
Skill	A procedure: steps, rules, output	Only when a request matches	Claude Skills (SKILL.md) · nearest in ChatGPT: a custom GPT
Connector	Access to an app or data source	When a task needs it	Connectors in both; many use MCP
Agent	A goal, tools and checkpoints	For a multi-step task	Claude Code · ChatGPT agent · Codex

Rule of thumb: tell a new research assistant once → context file. Hand them a checklist → skill.

Two plain-text files from today's conversation

research-context.md

Data

- Islam et al. (2026), CC BY 4.0

Coding

- career_anxiety: No=0 ... High=3

Ethics (EWU)

- No cells under 10; disclose AI

survey-analysis-workflow/SKILL.md

name: survey-analysis-workflow

description: Runs a verified survey analysis... Use when asked to analyse survey data.

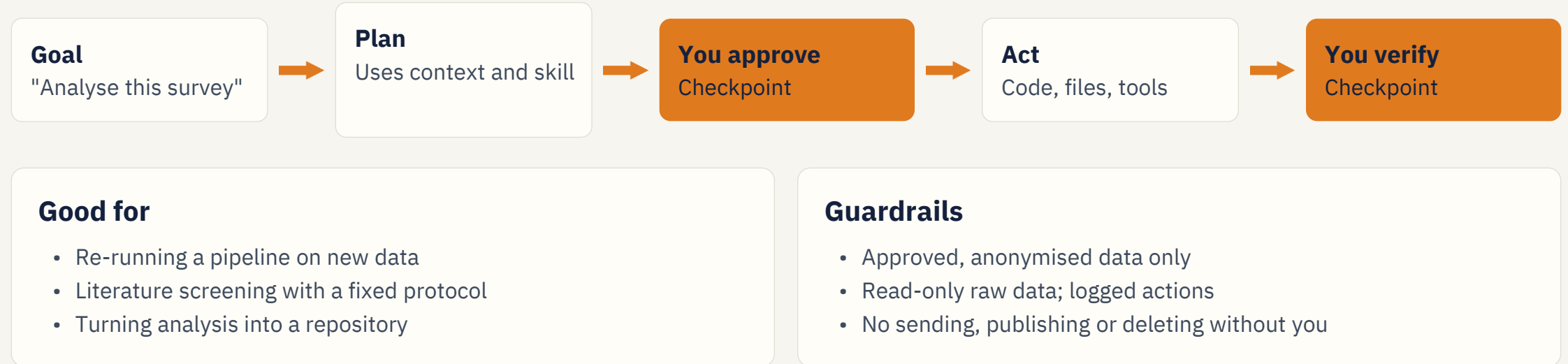
Steps

1. Ethics gate ... 9. Verify

Context file: paste into a Claude or ChatGPT Project's instructions.

Skill: zip the folder; upload in Claude at Customize > Skills.

An agent runs the whole workflow; you keep the gates



YOUR RESEARCH ECOSYSTEM

Seven layers, with you at the top

You and EWUREC	Questions, ethics approval, decisions, authorship
Agents	Multi-step pipelines with approval checkpoints
Tools and connectors	Code execution, Colab, Drive, GitHub, Zotero
Skills	survey-analysis, literature-screening, apa-reporting
Context files	research-context.md per study; CLAUDE.md in code folders
Projects	One per study: protocol, codebook, key papers
Data and records	Read-only raw data, data plan, prompt log, Git repository

THE IDEAL APPROACH

Build it over four weeks, one layer at a time

Week 1

Foundation

GitHub repository, read-only raw data, a prompt log

Week 2

Project and context

One Project per study with research-context.md

Week 3

First skill

Package the procedure you repeat most

Week 4

Tools, then agents

Connect Drive or GitHub; try an agent on a copy

Privacy by design: AI sees anonymised data only

Version everything: prompts, context, skills, code

Share to reproduce: others can rerun your work

AI drafts. You decide.

See the code

Every number comes from code you can read

Re-run it

Recompute before results leave the chat

Disclose it

Name the tool and what it did

TAKE-AWAYS

Standard prompts, real data, ethical by default

Ethics first

EWUREC for new data; anonymise; disclose.

Prompt with structure

Role, context, task, format, check.

Grow your ecosystem

Project, context file, skill, then agents.

Prompts, data, website and slides

github.com/rifat963/ewu-ai-training

Questions?